Quotation No : Q25000339

Date : 03-Sep-2025

Absstem Industrial Solutions Pvt. Ltd.
591
MANESAR
HARYANA

Kind Attention : Mr./Ms. Amit Sahni

Dear Sir,

We understand that in industry, reliability is not just a requirement,it is the foundation of success.
That is why Absstem Industrial PSA Generator is more than just a product; it is a commitment to excellence, efficiency, and uninterrupted performance.

Every component is designed with precision and innovation, ensuring that each molecule of oxygen and nitrogen is delivered with the utmost purity.

With fully automated operation, our system seamlessly integrates into your processes, reducing manual intervention and maximizing productivity.

Moreover, our industry-leading air ratio ensures that you achieve the highest efficiency, optimizing energy consumption while maintaining peak performance.
And because we believe in standing by our craftsmanship, we offer an unmatched-10 year warranty on sieves, giving you the peace of mind that your investment is protected for the long run.

We have attached the detailed quotation for your review.

Should you have any questions or require further customization to meet your specific needs, please do not hesitate to reach out.

We are here to provide not just a product but a solution that breathes life into your operations.

We look forward to the opportunity to serve you and build a lasting partnership.

Thanks and Regards,

For Absstem Industrial Solutions Pvt. Ltd.

Amit Sahni
Chief Executive Officer
9582499594
www.absstem.com



nGEN POWERED BY ABSSTEM INDUSTRIAL

Industrial

PSA Nitrogen Generator

Unique Features of Absstem Nitrogen Generator nGEN SH

a. **Lowest Air to Gas ratio** in industry thereby guaranteeing **lowest power consumption**.
b. **10 years*** warranty on Molecular Sieves **(First In Industry)**.
c. Assured Performance for consistent Flow, Pressure and Purity of Nitrogen throughout Product Lifecycle.
d. IIOT ready machine having 100% operator free design for 24X7 operation.
e. Idle mode with auto start/stop features ensuring power savings during low/no demand.
f. Inbuilt dew point transmitter to monitor input compressed air quality as standard scope of supply.
g. Plug n Play machine with an ergonomic design suitable for accessibility.
h. All the Instruments are specially manufactured to make them suitable for Nitrogen gas.
i. All critical performance parameters are being monitored through HMI.
j. First in industry **Comprehensive Maintenance Contract (CMC)** service under Absstem Premium Shield program providing complete peace of mind to customer.

Ambient Design Parameters

a.	Temperature	0 to 45 Deg. Celsius
b.	Relative Humidity	up to 80 % (Max.)
c.	Altitude	up to 1000 m above Sea level

Absstem Industrial Solutions Pvt. Ltd.
Plot No. 591, Sector 8, Phase IV
IMT Manesar, Gurugram, Haryana - 122051
Contact - 1800 3010 3394 Email - indsales@absstem.com
CIN: U29256HR2022PTC103387

Product Description

a. Product Name	High Purity PSA Nitrogen Gas Generator
b. PSA Module	**nGEN SH**
c. Purity Range	99% - 99.95%

General PSA Generation System Layout

A typical Industrial Nitrogen Generation system consists of three parts:

- Compressed Air System: Air Compressor, Pre-filter, Air Dryer, Dry Air Vessel
- PSA N_2 Generator
- Nitrogen Storage Tank



Air Compressor	An Air compressor is used to compress the atmospheric air to a pressure required for the separation of **Nitrogen** from the air.
Refrigeration dryer	Compressed air then passes through a refrigerated dryer to remove moisture (which is harmful to sieves) from the air.
Air Tank	Dry air is collected in an air tank to remove the pulsations which serve the dynamic (changing) requirement of air by the PSA system.
Multistage Filtration system	Before air enters the PSA system, it is passed through multistage filtration removing moisture, particulates and oil content to the specified levels.
PSA system	Pressurized air is passed through PSA Towers filled with Molecular sieves to separate **Nitrogen** from Air.
Storage Tank	Industrial Grade **Nitrogen** produced from the PSA system is stored in this tank for continuous supply.
Oxygen Analyser	Nitrogen produced is monitored by an Oxygen analyser. In case, O_2 impurity goes above the specified level, it gives an alarm and switches the plant to vent mode.

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HD Touch Screen Display

(User-friendly touch screen interface for easy operation and maintenance)

Absstem industrial Nitrogen generators complies with all industry standards and regulation

- Alarm management system
- Process data display
- Consumables running hours' display
- Data logging via USB drive**
- Graphical representation**



Additional upgrades

- Digital flow meter with totalizer
- Nitrogen Pressure monitoring for High pressure booster driven plants
- Cylinder manifold and alarm management system
- Remote monitoring
- Nitrogen gas Dew-point monitoring

International Quality standards

A strict quality procedure, following the leading international pharmacopoeia and Industrial directives, is followed in the selection of all mechanical and electrical instruments, vessel fabrication and designing of Industrial Gas generators.

TÜV NORD ISO 9001 · TÜV NORD ISO 14001 · TÜV NORD ISO 13485 · RIR CERTIFICATION Partners in Growth

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Absstem Shield Premium

(Comprehensive Annual Maintenance Contract)



Any Industry is now heavily focused on increased OEE (Overall Equipment Efficiency) and reduced LCC (Life Cycle Cost) while deciding to invest in any equipment. At Absstem, we strive for an everlasting engagement with our customers by not selling them just a product but also provide a long term after-sales support.

We understand that to deliver a consistent performance, periodic health check and timely maintenance of the Nitrogen Generator is extremely important. With Absstem Shield Premium, you invest in complete peace of mind for all maintenance needs pertaining to N2 generators and associated integrated equipment like air compressor, Air dryer etc.

Few benefits of Absstem Shield Premium are listed as below: -

1) Guaranteed assurance of plant performance.
2) Genuine spare parts and consumables being used to provide longevity and durability.
3) Regular visits by our technicians for health monitoring and carrying out the scheduled maintenance services.
4) 24 x 7 On-Call service support to our customers for any basic trouble shooting
5) Emergency visit to attend any unplanned breakdown to any location within 48 hours.
6) No additional surprise charges as all breakdown spares and consumables covered.
7) The contract can be signed at the time of equipment ordering itself and can be renewed on year on year basis.

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Product Description

Product	Description	Make	Qty
Gas Storage Vessel	AI-4000L Storage Vessel		1

Caption	Specification
Water Volume	4000 Litres
Gas Stored	Nitrogen/Oxygen
Design Standard	ASME Sec. VIII Div.1 Pressure Vessel Design
Working Pressure	8 Barg
Hydrotest Pressure	13.2 Barg
Accessories Included	Pressure Guage Safety Valve Manual Drain Additional Port for Other Instruments
Material of Construction	IS2062 Gr.B
Gas Stored Volume	26 M3
Corrosion Allowance	1.5 MM

Product	Description	Make	Qty
nGen Nitrogen Generator High Purity	nGEN090SH Nitrogen		1

Caption	Specification
Gas Generated	Nitrogen
Gas Pressure	upto 6.5 Barg
FLOW AT 99% PURITY	254 Nm3/Hr DRY AIR REQUIRED 329 CFM
FLOW AT 99.5% PURITY	212 Nm3/Hr DRY AIR REQUIRED 318 CFM
FLOW AT 99.9% PURITY	158 Nm3/Hr DRY AIR REQUIRED 299 CFM
FLOW AT 99.95% PURITY	142 Nm3/Hr DRY AIR REQUIRED 313 CFM
COMPRESSED AIR PRESSURE REQUIRED	7.5 Barg
RECOMMENDED NITROGEN STORAGE	4000 Liters
POWER SUPPLY	230 V/ 110 V 50/60 Hz Single Phase
RECOMMENDED AIR RECEIVER	3000 Litres Part of Air System
INPUT AIR DEW POINT MONITOR	INCLUDED
CMS Warranty	10 Years*

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Product	Description	Make	Qty
Compressed Air System	C045F0275 Air System		1

Caption	Specification
Air Compressor KW	45 KW
Air Compressor Make	Atlas Copco/ Ingersoll Rand/ Gardner Denver/ As per Demand
Air Dryer	Included
Dry Air Storage	2500 Liters Included
Interconnecting Piping	Included
Pre filter	Included
Dry Air Generated	275 CFM
Air Pressure	7.5 Barg
POWER SUPPLY	400 V 3 Phase 50 Hz

Commercial Offer

S No	Item Code	Item Name	Description	HSN	Qty	Unit Price	GST(%)	Amount
1	AIS0408ST	AI-4000L	Gas Storage Vessel	731100	1	2,80,000.00	18.00	2,80,000.00
2	nGEN090SH	nGEN090SH	nGen Nitrogen Generator High Purity	840510	1	46,77,000.00	18.00	46,77,000.00
3	ACSC045F	C045F0275	Compressed Air System	840510	1	20,90,000.00	18.00	20,90,000.00
							Sub Total	70,47,000.00
							GST(%) @18.00	12,68,460.00
							Grand Total ()	83,15,460.00

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Terms And Conditions

1.	EXCLUSIONS & CLIENTS SCOPE	All erection / civil work, All Power Cables & Cabling out of battery limit of PSA Plant, Cable Trenches etc. Unloading / Shifting of Gas Plant at Clients site will be in Client Scope. Additional gas piping, cooling water (if required).
2.	SUPERVISION OF ERECTION & COMMISSIONING	There shall be no commissioning charges, however you shall arrange freeTo & Fro Air Ticket, lodging / boarding, and local conveyance for our Engineers during supervision of Installation of Gas Plant at site on free of cost basis. Our Engineers will be there for Supervision only. Any manpower / fitter required for installation work shall be provided by you.
3.	P&F	3% Extra.
4.	Freight	Extra at Actuals.
5.	Insurance	In your Scope.
6.	Payment Terms	50% Advance with your Prestegious Purchase Order and Balance against PI.
7.	Delivery	6-8 Weeks from the PO and Advance. Rest can be confirmed for Availability.
8.	Warranty	We hereby confirm that, our equipment shall have Suppliers warranty for a period of 12 months from the date of Commissioning or 18 months from date of Dispatch whichever is earlier.
9.	GST	As per the Commercial Table.
10.	Remarks	.

Hope you will find our offer in line with your requirement.

Should you require any further clarification/information, please feel free to contact us.

Thanking you, and assuring you of our best attention at all times.

Very Yours truly,

For Absstem Industrial Solutions Pvt. Ltd.

Amit Sahni

Chief Executive Officer

9582499594

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Savings Calculator

Nitrogen Gas Supply Cost Comparison



Annual Savings Summary

₹ 4,32,97,536

Estimated annual savings with PSA system

Input Parameters

Cylinders used/day

192 No.

Plant running hours/day

12 hr.

Plant running days/month

25 days

Nitrogen Purity

Built with JDoodle.ai

99.9 %

Cylinder Volume

10 m^3

Cost of 1 Cylinder

750 Rs

Rental Cost/Month (if any)

Rs

PSA Plant Flow

186 Nm^3/hr

Compressor Power

55 KW

Load Factor

0.9

Power Cost/Unit

8 Rs.

Cylinder Operator Cost/Month

24000 Rs.

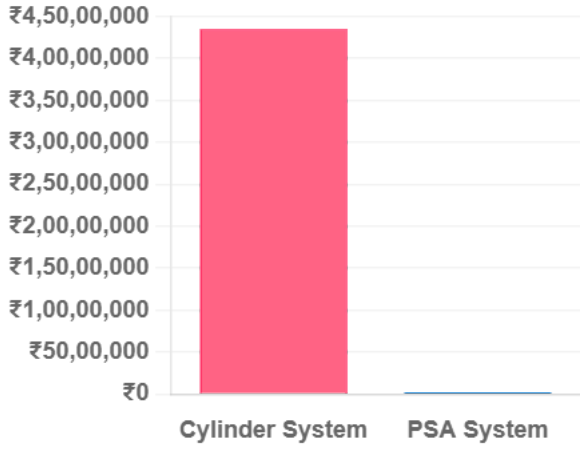
PSA Operator Cost/Month

12000 Rs.

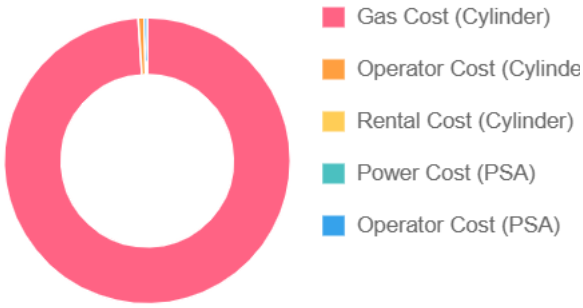
Cost Analysis Visualization

Built with JDoodle.ai

Annual Operating Cost Comparison



Cost Breakdown Analysis



Built with JDoodle.ai

Cost Comparison Results

Per hour consumption of gas	160 Nm^3/hr	Air Requirement	352 CFM
Unit Price / M^3	75 Rs	Present Consumption / Month	48,000 Nm^3
Total Expense / Month	3,600,000 Rs	Annual Rental Cost (if any)	0 Rs
Power	49.5 KW	Compressor Utilization Factor	97.8%
Unit Price / M^3 in PSA	2.08 Rs	Saving/m^3	72.92 Rs
Monthly saving by using PSA	35,00,077 Rs	Yearly saving by using PSA	4,20,00,929 Rs
Operator cost/year (cylinder)	288,000 Rs	Operator cost/year in PSA	144,000 Rs
Total running cost in Cylinders / year	43,488,000 Rs	Total running cost in PSA/year	190,464 Rs
Annual saving	4,32,97,536 Rs		

absstem
REINVENTING PERFORMANCE

This savings calculation is provided as an estimate. Actual savings may vary based on operating conditions.

Report generated on 9/10/2025

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